

Response to the Editor and Reviewers

Climate outweighs human effects on pollen assemblage dynamics from 8.5 to 2.0 ka BP

EDITORS REQUESTS

In particular the following editorial thresholds needs to be met,

- Provide a compelling case that climate outweighs human effects on vegetation assemblage dynamics in the Holocene, including a thorough analysis of the results, clearly indicating the variance explained by models.

Response. We have strengthened the evidence for the central conclusion and clarified what the models do and do not show. The revised article now distinguishes the overall explanatory power of each model from the relative contributions associated with palaeoclimate and human activity. During revision, we also corrected the extraction of these contributions and examined whether the conclusion depended on alternative treatments of shared or negative components. Palaeoclimate had the larger estimated contribution in 891 of the 1,098 pollen records in the controlled spatial analysis (81.1%). For these same records, the mean adjusted R^2 was 0.440 and the median was 0.450. The regional analysis through time gave the same general result: climate had the larger estimated contribution in 69 of 70 comparisons for which both climate and archaeological evidence could be evaluated.

Changes in the revised article. These results are reported in “H1: Holocene human impact secondary to palaeoclimate” and in Figs. 3–4. The Supplementary Information documents model fit, uncertainty arising from shared explanatory variation, and alternative ways of summarising the contributions. The Discussion now makes clear that these contributions divide the variation explained by a model; they do not represent proportions of all ecological change recorded in the pollen data.

- Justify the usage of data from multiple scales and fully discuss the limitations of your methods, particularly as raised by reviewer #2.

Response. We have revised the article to describe explicitly the different spatial worlds represented by the evidence. A pollen sequence reflects vegetation within a source area that differs among lakes, bogs, and depositional settings. The climatic values describe conditions

at the location of each record, whereas the archaeological SPDs draw on radiocarbon dates from a much broader surrounding landscape. The expert chronologies represent regionally informed interpretations of changes within individual pollen diagrams. These sources cannot be made spatially equivalent within a global compilation. We therefore restrict our conclusion to continental-group and regional patterns between 8.5 and 2.0 ka BP and discuss how scale affects ecological interpretation. We also compare results obtained with 250 and 500 km archaeological neighbourhoods and with alternative evidence for human activity. The revised Discussion gives fuller attention to pollen production and count size, regional differences in archaeological knowledge, climatic smoothing, shifting Holocene climates, and the persistence of spatial and temporal structure in ecological records.

Changes in the revised article. The spatial meaning and limitations of each source are now explained in the Methods sections on data filtering, spatial units, human activity, palaeoclimate, and H1. Their consequences for interpreting local and regional ecological change are developed in the Discussion, with the supporting analyses reported in the Supplementary Information.

- Clearly explain your methods in detail.

Response. We have expanded the Methods so that readers can follow the evidence from pollen counts and archaeological dates to the ecological comparison presented in the Results. The revised text explains how records were selected, how the ten pollen assemblage properties were calculated, how temporal trajectories were summarised without allowing long or densely sampled records to dominate, and how the climate and human-history variables were assembled. It also distinguishes the amount of variation explained by a model from the division of that explained variation between climate and human activity. The treatment of time, spatial pattern, and the supplementary robustness analyses is now stated explicitly. The guide below shows which pollen records or continental-group time intervals contribute to each part of the study.

Changes in the revised article. These explanations are integrated throughout the Methods. More specialised checks are collected in “Supplementary Methods: Result-validation analyses,” and each figure and table now states the records, regions, or time intervals on which it is based.

Cohort guide

The rows below refer to different bodies of evidence and, in two cases, different units of observation. They are therefore not successive losses from a single dataset. Full definitions are provided in Supplementary Table S1.

Evidence used	Count	Records or comparisons represented
Prepared pollen collection	1,270	Pollen records after baseline preparation
Records considered for the original H1 spatial analysis	1,262	Records represented in the original model inventory
Records with an estimable original H1 comparison	1,156	Records for which climate and human contributions could be estimated
Principal spatial analysis	1,098	Pollen records analysed after accounting for time within each sequence
Principal regional analysis through time	70	Regional time intervals analysed after accounting for geographical pattern
Spatial comparison of archaeological and expert evidence	151	Records with both archaeological and expert evidence available for comparison
Temporal comparison of archaeological and expert evidence	61	Regional time intervals with both forms of human evidence available

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author)

Review of the manuscript

“Climate outweighs human effects on vegetation assemblage dynamics in the Holocene” by Felde et al.

The manuscript submitted by Felde et al. presents a substantial contribution to the field of paleoecology. The study is based on a large dataset of pollen records spanning all continents to investigate the relative influence of climate and human activity on vegetation compositional changes throughout the Holocene. The authors use modelled climate data and archaeological radiocarbon dates, along with expert knowledge on human activities derived from pollen records, to disentangle the effects of climate and human activities on vegetation/pollen composition and diversity. The analytical framework is well developed, employing state-of-the-art statistical methods to explore large-scale pollen composition and diversity changes. The use of Pollen Assemblage Properties (PAPs) to assess compositional changes is innovative and adds a new

dimension to paleoecological interpretation. Overall, this is an important and ambitious manuscript that I highly recommend for publication.

My major comments on the manuscript are listed below.

Response. We thank the reviewer for the constructive assessment. The revision defines more carefully the ecological meaning of the pollen assemblage properties, the historical interval and spatial scales to which the conclusions apply, and the strengths and limitations of the climate and archaeological evidence.

Scope and framing of the study period

The title suggests an analysis of the entire Holocene, yet the dataset ends approximately 2000 years ago. This period excludes the last two millennia, during which anthropogenic impacts on vegetation intensified, including major deforestation events and agricultural expansion. The title should be revised to more accurately reflect the temporal scope of the data and findings. Furthermore, some statements in the discussion, such as the caution against the narrative that human transformation is always destructive, may appear strong given the temporal cut-off at 2 ka BP. I suggest to soften or contextualize this statement to avoid misinterpretation.

Response. We agree. The title now specifies the interval 8.5–2.0 ka BP and refers to pollen assemblage dynamics. This is the period for which the archaeological SPD evidence supports the principal comparison. Some expert-identified events occur after 2.0 ka BP, but these younger observations are shown separately and do not extend the principal conclusion into the last two millennia. We have also removed the broad statement that human transformation was not destructive. The revised Discussion instead emphasises that our result concerns the spatially aggregated ecological patterns represented in this study and does not diminish the extensive historical and palaeoecological evidence for substantial local human transformation.

Changes in the revised article. The title is now “Climate outweighs human effects on pollen assemblage dynamics from 8.5 to 2.0 ka BP.” The same temporal and ecological scope is maintained in the Abstract, hypotheses, Results, Discussion, and Conclusion. Younger event observations are identified separately in Fig. 4.

Use and justification of climate model outputs and human activity proxies

While the use of paleoclimate model data is justified by the limited spatial coverage of proxy-based climate reconstructions, a brief discussion of the benefits and limitations of using model outputs (e.g., smoothed trends and difficulties to simulate climate in some regions) should be included. This will help the readers to better understand the implications using climate model outputs.

Response. We agree and have added a fuller discussion. CHELSA-TraCE21k provides a spatially continuous and internally consistent description of climatic change across regions where proxy reconstructions remain uneven. This is valuable for a global comparison, but the simulations smooth short-lived and local climatic variation, depend on model assumptions, and may not reproduce conditions within the actual pollen source area, especially where elevation or topography changes over short distances. We therefore interpret the climate variables as broad representations of climatic history rather than exact reconstructions of the environments surrounding individual sites.

Changes in the revised article. The advantages and limitations of the simulated climate histories are now explained under “Palaeoclimate” and revisited in the Discussion when comparing local ecological change with broad climatic patterns.

In a same way, the use of summed probability densities (SPDs) from archaeological radiocarbon dates is appropriate but should be more critically discussed, although the authors already mention some of the issues. These can underestimate human presence, especially in regions or periods with sparse data. Furthermore, the expert-derived “human-impact events” from pollen data introduce a degree of circular reasoning as it is based on pollen information. The authors should mention the pros and cons of using such approach.

Response. We agree. We retain archaeological SPDs as validated proxies for regional human population changes, while making clear that they are histories of dated archaeological activity rather than direct censuses of past populations. Their form is influenced by preservation, calibration, archaeological research traditions, and uneven knowledge among regions and periods. SPD remains the principal human-history proxy because it is independent of the pollen response and can be compared across much of the study area. The expert-derived events contribute a different kind of evidence: regionally informed indications of land use and vegetation disturbance within pollen diagrams. They can reveal changes not represented by a regional SPD, but they are unevenly available and partly circular because both the event and the ecological response may be inferred from the same pollen sequence. We therefore use them to examine sensitivity to the choice of historical evidence, not as independent confirmation.

Changes in the revised article. These complementary strengths and limitations are now explained in the Methods and Discussion. The Supplementary Information shows how the results change when archaeological SPDs, expert-derived events, or both are used to represent human activity.

Background information and previous studies

The introduction would benefit from presenting a bit more prior studies that used similar approaches, e.g., Kuosmanen et al. (2018), Marquer et al. (2017), and Reitalu et al. (2015). These

studies provide an important context for methods, data types, and interpretations of vegetation change drivers, although for Europe and regional scales only. Their inclusion would strengthen the rationale for the current study.

Response. We agree that the regional literature is essential context. The Introduction and Discussion now relate our study to European and other regional attempts to separate climatic and human influences on past vegetation. We also clarify an important ecological distinction: our response variables describe richness, evenness, turnover, and other properties of pollen assemblages, whereas several earlier studies reconstructed changes in particular taxa or vegetation composition.

Changes in the revised article. The revised text discusses Kuosmanen et al. (2018), Marquer et al. (2017), Reitalu et al. (2013) on climatic and human influences on forest composition, and Reitalu et al. (2015) on functional and phylogenetic diversity inferred from pollen.

Interpretation of hypotheses and results

Hypothesis 1 states that if human impact is an important driver, its influence should match or exceed that of climate. I would say “the major driver” rather than “important driver” in the case it exceeds climate.

Response. We agree and now describe human activity as the proposed “primary driver” in H1. This wording reflects the intended comparison with climate without implying that a smaller estimated contribution would make human influence ecologically unimportant.

Changes in the revised article. H1 now reads: “The first hypothesis (H1) proposed that human impact was the primary driver of changes in PAPs at continental-group and regional scales between 8.5 and 2.0 ka BP.”

The authors conclude that climate dominates vegetation change at large spatial scales rather than local scales. This aligns with prior studies. I agree with the authors and clearly support the fact that at finer spatial resolutions, human impacts become more visible and potentially dominant. It is important to mention that the use of PAPs may homogenize vegetation signals and amplify climate-related patterns. This should be acknowledged.

Response. We agree and now distinguish the geographical coherence of a driver from the ecological importance of its local effects. Climatic change can produce related responses across widely separated pollen records, whereas cultivation, burning, grazing, woodland clearance, and settlement often have intense but spatially heterogeneous effects. Summarising pollen assemblage properties across continental-group and regional units may therefore accentuate the coherent climatic component while obscuring diverse local histories of land use. The revised interpretation makes this scale dependence explicit.

Changes in the revised article. The Discussion now cautions that the broad climatic pattern should not be used to discount well-supported local histories of human transformation. It also explains how the spatial extent of pollen recruitment, archaeological evidence, and analytical regions affects the comparison.

The manuscript would be greatly improved if the authors provide a discussion of vegetation changes themselves, and how specific taxa or assemblage types respond to climatic or anthropogenic drivers. Such discussion would help ecological interpretations.

Response. We agree that the ecological interpretation required expansion. Richness records how many pollen types were detected; evenness describes how evenly they were represented; turnover and rate of change indicate the degree to which assemblages were reorganised through time. These properties can reflect migration, community reassembly, changes in dominance, disturbance, and land use. They are also shaped by pollen productivity and dispersal, basin setting, taxonomic resolution, and sampling effort. We therefore interpret them as properties of the pollen assemblage and avoid attributing a climatic or human contribution to particular trees, grasses, crops, or disturbance indicators without a taxon-level analysis.

Changes in the revised article. The Discussion now explains the ecological meaning of each family of pollen assemblage properties and distinguishes this approach from reconstructions of particular taxa or land-cover classes.

Data representativeness and site characteristics

The choice of 1262 pollen sequences is impressive, but more information is needed about the selection criteria. I would have assumed that much more sites would be available at such spatial scales.

Response. We have expanded the account of how the pollen archive was assembled. Duplicate sequences were removed; samples outside the study interval or lacking adequate chronological control were excluded; each retained sample contained at least 25 terrestrial pollen grains; and each sequence contained at least five usable samples. Taxonomic names were harmonised before the assemblage properties were calculated. Africa was not included because the available records were too sparse when the compilation was assembled to support comparable regional analyses. This procedure yielded 1,270 prepared records, of which 1,262 entered the original H1 spatial analysis. Particular analyses use fewer records where the necessary climate, archaeological, ecological, or control information is absent.

Changes in the revised article. The full selection criteria are now listed under “Data filtering.” Supplementary Table S1 and the guide at the beginning of this letter identify the records or regional comparisons represented in each analysis. Because the original search

did not retain a complete count of every potentially available public record, we do not present the prepared collection as a fully quantified global census of pollen sequences.

Site characteristics (e.g., basin size, proximity to human settlements) can significantly influence pollen source area and the sensitivity to land use signals. This aspect, although mentioned by the authors, should be discussed a bit more in details, especially when interpreting local vs regional patterns.

Response. We agree. Basin size and depositional setting determine how much of the surrounding vegetation contributes pollen to a sequence, while distance from settlements or cultivated land affects the likelihood that local land use will be recorded. These differences are ecologically important when a site-scale pollen signal is compared with an archaeological history assembled over hundreds of kilometres. Consistent information on basin geometry, depositional environment, and settlement proximity is unavailable for much of the global compilation, so these factors could not be represented uniformly in the models. We consequently restrict the interpretation to broad spatial patterns and discuss why individual sites may depart from them.

Changes in the revised article. The Methods and Discussion now give greater attention to basin setting, pollen source area, and the contrast between local land-use histories and broad archaeological neighbourhoods. A separate comparison shows how the results change when dates are drawn from 250 rather than 500 km around a pollen site.

The analysis assumes stable spatial climate zones. However, climate zones have shifted over the Holocene, which could affect interpretations of vegetation-climate relationships. A brief discussion of this dynamic would be valuable.

Response. We agree that climatic boundaries moved during the Holocene. We now call the five broad geographical units *continental groups* and their smaller analytical subdivisions *regions*. The regions were defined reproducibly from present-day Köppen–Geiger classes to divide the records into groups that are easier to compare. This is a practical classification, not a claim that these are natural, permanent, or uniquely correct ecological boundaries. A site assigned to a region using its modern Köppen–Geiger class may have experienced different climatic conditions in the past. The continuous palaeoclimate histories, rather than the fixed regional assignments, provide the climatic evidence in the fitted models.

Changes in the revised article. The Methods now define *continental groups* and *regions* and explain why the regions were formed. Figure captions, tables, and the Discussion use these terms consistently and state the limits of a fixed modern classification.

Thresholds for pollen counts

The decision to lower the minimum pollen count threshold from 150 to 25 grains per sample raises some questions about data quality. A threshold as low as 25 may introduce significant noise and reduce the reliability of PAPs. The rationale for this decision should be clearly explained and its implications discussed.

Response. We agree that small pollen counts make richness, evenness, and turnover less precise. We have clarified that 25 grains is the minimum for retaining an individual sample, whereas 150 grains is the usual rarefaction target. A lower record-specific target is used only when a sequence cannot consistently support 150 grains. Retaining these records preserves evidence from regions where pollen archives are sparse. We also quantified the geographical cost of more demanding targets. A target of 300 grains retains 1105 of 1270 records; a target of 500 grains retains 616; and a target of 1,000 grains retains only 246. The losses are strongly uneven among regions and periods, so stricter filtering would narrow both the quantity and historical representativeness of the evidence. These comparisons raise the rarefaction target and its associated minimum; they do not require every sample to contain 300 or 500 grains, and they do not constitute a refitted H1 analysis at those strict minima.

Changes in the revised article. The Methods now distinguish the minimum count, the usual rarefaction target, and the lower targets used for individual records. Supplementary Fig. S28 and Supplementary Table S3 show how stricter targets reduce regional and temporal coverage, and the Discussion treats finite counts and pollen production as limitations on ecological interpretation.

Figures

Figure 4 shows a lot of information, and the plots are too small. It is difficult to see the major patterns. Furthermore, the panel labels in Figure 2 appear to be misaligned (E and D).

Response. We agree and have reorganised the figures around the ecological sequence of the argument. Fig. 1 introduces the study design. Fig. 2 then shows six representative histories of pollen assemblage change, with all ten properties presented in Supplementary Figs. S6–S15. Figs. 3 and 4 compare the climatic and human contributions in space and through time. The former H2 figure is now Supplementary Fig. S1, and the remaining robustness analyses are grouped in the Supplementary Information.

Changes in the revised article. Figs. 2–4 now each address one principal question at a readable scale, and their captions identify the evidence, temporal or spatial controls, and treatment of uncertain estimates.

This manuscript provides a novel and valuable analysis of Holocene vegetation dynamics at broad spatial scales. The conclusions are well supported, and my major comments focus on

some aspects of the methods and interpretation that require clarification and additional nuance. By addressing these issues, the manuscript will be greatly improved. Again, I recommend publication after these revisions are done.

Response. We thank the reviewer again. The revised article now places the statistical comparison within a clearer account of the pollen archive, the historical proxies, the spatial scale of ecological change, and the limits of broad-scale inference.

Reviewer #2 (Remarks to the Author)

This paper uses a global dataset of fossil pollen, paleoclimate, archaeologically-attested radiocarbon dates and human indicator pollen types to investigate the relative importance of humans and climate on aspects of pollen diversity and compositional changes from 8.5 cal k BP onwards. Previous work has described changes to various aspects of pollen diversity and compositional turnover measures over multi-millennial timescales across much of the earth surface. Other studies have investigated the relative strengths of relationships between humans versus climate on pollen diversity and compositional turnover for different continents and regions. However, this study represents the first attempt to quantify the relative importance of humans and climate on pollen diversity and compositional turnover measures at a global scale in the Holocene. Using a new method to the field of paleoecology, the authors claim that, at continental and regional scales, climate was the primary driver of pollen diversity and compositional turnover measures over the period 8 - 2.5 cal k BP. This work adds a novel dimension to the conversation (a direct comparison of the strength of drivers globally) and thus represents an interesting and worthwhile contribution to the literature.

Response. We thank the reviewer for recognising the scope of the study and for identifying where the ecological evidence and its limitations required clearer explanation. We now refer consistently to pollen assemblage dynamics, define 8.5–2.0 ka BP as the interval supported by the principal archaeological comparison, and show younger expert-derived events separately. Histories of the pollen assemblage properties now precede the comparison of climate and human activity, and the revised Methods and Discussion give fuller attention to the construction, spatial meaning, and limitations of each source of evidence.

I do, however, have a number of data presentation and methodological concerns that I would need to see addressed to have confidence in the results presented here. I also see no advantage of including the human event variable here. I understand that the authors would prefer to be able to run these analyses up to the present, but I find the variable reductive and without nuance, it is un-tested and it is much too coarse for it to ever be likely to capture changes to local PAPs. In addition to methodological considerations, I would also like to see an adjustment to the title. In its current form, at least to me, the title suggests that this is a study encompassing the full Holocene, though the primary analyses span the period 8.5 - 2 cal k BP. This is very important because recent work has shown substantial increases in multiple measures of pollen diversity from 2.5 cal k BP onwards across the planet (e.g. in Europe, Giesecke et al, 2019; North America,

Stegner and Spanbauer, 2023; global, Mottl et al, 2021 and Gordon et al, 2024), a period of known rapid and major change in human population size, culture, etc. These changes are missed in these analyses given the cut-off date. Climate outweighs human effects on vegetation assemblage dynamics 8 - 2.5 ka BP (or similar) would seem to me to be a more accurate and less ambiguous representation of the analyses.

Response. We agree that the former title overstated both the temporal extent and the ecological domain of the study. The revised title is “Climate outweighs human effects on pollen assemblage dynamics from 8.5 to 2.0 ka BP,” and the conclusion is explicitly restricted to this interval. The expert event chronology is too coarse to represent the full diversity of local human activity, but we have retained it as secondary historical evidence. It records regionally interpreted indications of disturbance and land use that may be absent from archaeological SPDs and, in some regions, extends into younger periods. Comparing the two sources shows whether the broad conclusion depends on how human activity is represented. SPD remains the principal proxy, and the event chronology is not treated as independent confirmation because it partly derives from the same pollen archive as the ecological response.

Changes in the revised article. The title, Abstract, hypotheses, Results, Discussion, and Conclusion now define the principal interval as 8.5–2.0 ka BP. Younger event observations are labelled separately in Fig. 4. The Methods and Supplementary Information explain the different historical evidence represented by SPDs and expert-interpreted pollen events.

Data presentation

Though two records are presented in Extended Fig. 3, a key figure missing from this work is a presentation of the full set of PAPs through time for each region. The authors have presented the set of predictor variables, rather than the response variables, which, in my view, are of substantially less importance than the response variables themselves. I find it difficult to make sense of and interpret statistical analyses for which at least some abstraction of the response variables (here, PAPs) are not presented in a paper and thus, for me, these need to be included.

Response. We agree. Readers should first see how the pollen assemblages changed before considering why they changed. Fig. 2 therefore presents six representative histories of richness, evenness, turnover, and related properties, while Supplementary Figs. S6–S15 show all ten properties individually. Each curve gives equal weight to the contributing pollen records and is evaluated only across the portion of time actually represented by each sequence, preventing long or densely sampled records from dominating the regional history.

Changes in the revised article. The Results now begin with “Temporal patterns in pollen assemblage properties.” Fig. 2 presents the principal ecological trajectories, and the

full set is shown in the Supplementary Information.

Pollen diversity biases

Whether implicitly or explicitly stated, pollen data in studies such as these are used to make inferences on the changing diversity patterns, and relationships with drivers, of the parent vegetative assemblages of pollen samples (as stated by the authors, L652). Thus, the treatment of the pollen data in order to make robust inferences about the actual focus – the parent vegetative assemblage – is of paramount importance. For the reasons outlined below, the treatment of the pollen data in these analyses means that I find it difficult to have full confidence in the robustness of the computed PAPs and thus confidence in the conclusions of this work.

There is substantial variation in pollen productivity among pollen producing plants, which has the capacity to strongly bias estimates of individual PAPs, which the authors acknowledge (L687). Pollen grains from high pollen producing plants are more likely to be deposited and identified in pollen samples than pollen grains from low pollen producing plants. The result is that at low pollen sums, pollen assemblages (samples) are likely to be swamped by pollen grains from a relatively smaller number of high pollen producing plants. This means that the evenness of these samples will be low (high representation by a small number of types), the result of which is a reduced likelihood of detecting pollen types from other pollen producers, which thus artificially depresses the richness of these samples (e.g. Giesecke et al, 2014). Thus at low pollen sums, the richness of samples primarily reflects the evenness of those samples. Similarly, if the likelihood of sampling a diverse set of grains per sample is reduced (given the low evenness of samples), compositional turnover estimates are likely to be lower for these samples than between samples with higher evenness values. In these analyses, pollen samples are rarefied to either 150 or 25 pollen grains. Even 150 grains is substantially lower than the minimum rarefaction value of ~300-500 grains commonly used in pollen diversity analyses. For those samples rarefied to 25 grains, diversity estimates will almost certainly be affected strongly by these confounding issues. Extended Fig. 3 describes the PAPs and predictor variables for two records, where these described issues can clearly be seen: those Hill numbers N0, N1 and N2 (~richness, ~diversity and ~evenness) are strongly correlated.

These concerns can be at least partly mitigated against by performing some transformation of pollen counts based on relative pollen production and dispersal (etc). There are a number of methods to achieve this, the most notable of which is the REVEALS model, for which there are the requisite datasets for many of the regions used here. However, this is not implemented in these analyses. At the very minimum, I would expect to see a supplementary analysis that considers only the subset of samples and records with a substantially higher minimum pollen grain N (300-500 grains), to check that results are not driven by those issues relating to relative differences in pollen production and dispersal described above. I am of the view that all samples with <150 grains should be excluded from these analyses.

Response. We agree that pollen productivity, dispersal, taxonomic resolution, basin setting, and count size all influence the ecological meaning of pollen assemblage properties. The article now refers to pollen assemblage dynamics rather than treating the measurements as direct estimates of plant abundance or vegetation diversity. We also clarify that 25 grains is the minimum for retaining an individual sample, not the count to which all samples are reduced. The usual rarefaction target is 150 grains; only sequences that cannot consistently support this target are analysed at a lower, record-specific count.

We examined how stronger count requirements would alter the geographical and temporal evidence available to the study. A 300-grain target, with a 50-grain sample minimum, retains 1105 records. A 500-grain target, with a sample minimum of about 83 grains, retains 616; a 1,000-grain target, with a sample minimum of about 167 grains, retains only 246. Thus the 500-grain target removes 51.5% of the baseline records and the 1,000-grain target removes 80.6%. These losses are not geographically or historically neutral: they disproportionately remove evidence from particular continental groups, regions, and periods (Supplementary Fig. S28; Supplementary Table S3). We therefore retain the baseline rule to preserve the broadest defensible coverage. This analysis does not answer the reviewer's proposed robustness test in full: we did not refit H1 after requiring 300–500 grains in every sample, and we do not claim that the estimated contributions would be unchanged under that restriction.

Richness, evenness, and the Hill-number measures are necessarily related because they describe different aspects of the same assemblage. We now show these relationships directly and repeat the comparison after removing the most closely related response measures. Across 1,114 records represented in both analyses, the median climatic contribution changed from 0.798 to 0.794. The broad climatic predominance remained, although the contribution estimated for individual pollen records sometimes changed.

We did not apply REVEALS because the pollen-productivity estimates, dispersal parameters, basin descriptions, and taxonomic resolution needed for a comparable analysis are not available across all regions. Restricting that correction to Europe or other well-studied regions would make the ecological meaning of the response differ among continental groups. The revised article therefore treats productivity and dispersal as unresolved limitations and does not imply that rarefaction corrects them.

Changes in the revised article. The count procedure and the relationships among pollen assemblage properties are now explained in the Methods. The Supplementary Information shows the loss of geographical and temporal coverage under stricter targets, the observed correlations among properties, and the reduced-response comparison. The Discussion contains a dedicated account of how pollen production, dispersal, and finite counts constrain ecological interpretation.

SPD biases

SPDs of archaeologically attested radiocarbon dates are becoming increasingly used to infer relative changes in human paleodemography. However, there is substantial and ongoing debate in the literature among archaeologists about whether this method can actually tell us anything about paleodemography (e.g. Chaput and Gajewski, 2016; Crema and Bevan, 2020; Crema, 2022). It is only material from times and places of archaeological interest that are radiocarbon dated. Thus the number of radiocarbon dated objects through time and space is intensely non-random, clustering around locations and periods of interest to the archaeological community. This is likely the cause of the substantial apparent paleodemographic spikes during transitions between archaeological periods (archaeologists are interested in what caused and was affected by these transitions). On top of this, radiocarbon dating is expensive. So, when possible, other objects – e.g., pottery, coins – are preferentially used to date archaeological remains, which results in a reduction in the number of radiocarbon dates when these objects are readily available. This is most obviously shown in Europe from ~3 cal k BP, from which point the radiocarbon SPD shows a monotonic decline (Extended Data Fig. 4). This apparent reduction is entirely artificial and driven by the general cessation of the use of radiocarbon dating to date archaeological remains, given the proliferation of other dateable material across the continent from this period onwards. Given that this variable underpins this paper’s primary analyses and main conclusions, I would expect some critical discussion in the main text that outlines the well-known and strongly debated limitations of using SPDs for paleodemographic inference, though this is missing. This would then help justify the decision to cut off the analyses at 2.5 cal k BP, which at the moment does not feel sufficiently justified in the text.

Response. We retain archaeological SPDs as validated proxies for regional human population changes, while agreeing that they are not direct population censuses. They are shaped by the survival and discovery of archaeological material, the history and intensity of research, laboratory dating practices, and the structure of the radiocarbon calibration curve. These influences vary among continental groups, regions, and periods and become particularly important near the younger boundary of this study. The regional distributions of dates and SPDs showed that 2.0 ka BP is the youngest common 500-year boundary at which a broad comparison remains supportable. We use it as a practical limit for this global historical comparison, not as a universal date after which SPDs cease to be informative.

Changes in the revised article. The Methods and Discussion now describe SPDs as archaeological histories whose interpretation depends on preservation and research practice. Supplementary Fig. S3 shows the regional coverage that informed the 2.0 ka BP boundary, and the article does not extend the principal conclusion into younger periods.

Notwithstanding these comments, of most concern to me is the different spatial scales of the respective variables used in these analyses. The relevant source area of pollen for any given extracted fossil pollen sequence varies from meters up to tens of kilometers (rarely hundreds

of km), depending on the deposition environment, location, etc. However, the authors pull radiocarbon dates from either 250km or 500km surrounding a pollen record extraction location to build their SPDs. These scales are vastly different and it is perhaps not surprising that PAPs from, for example, a bog site with a pollen source area in the order of hundreds of meters are not associated with SPDs built from radiocarbon dates pulled from within 250 or 500km but that they are associated with paleoclimate estimates for the actual location of a pollen record. In principle, either (i) a dynamic radius for retrieving radiocarbon dates per pollen record could be used (based on each record's pollen source area), (ii) these analyses could be run regionally (using averages across PAPs and predictor variables) or (iii), at the very least, a sensitivity analysis should be run that restricts the analyses to those records with the largest pollen source areas, with SPDs built using a radius for radiocarbon dates that is at least equal in order of magnitude to it.

Response. We agree that the pollen and archaeological evidence describe landscapes at very different scales. Pollen deposited in a small bog may chiefly reflect vegetation within a few kilometres or less, whereas an SPD assembled within 250 or 500 km describes a much broader archaeological landscape. Ideally, the archaeological neighbourhood would be tailored to the source area of each pollen record. This is not possible here because basin dimensions, depositional settings, and estimates of pollen source area are unavailable or inconsistent for much of the global compilation. We therefore compared the same pollen records using archaeological dates drawn from 250 and 500 km. Among 999 records available for both comparisons, 94 changed from a climate-dominant to a human-dominant result, or vice versa; among 70 continental-group time comparisons, 3 changed ordering. The median change in the spatial climate–human balance was 0.025. This shows that the chosen archaeological neighbourhood affects some individual results, while leaving the broad pattern intact. Because even 250 km greatly exceeds the pollen source area of many sites, this comparison does not remove the mismatch. We have accordingly restricted the conclusion to broad regional patterns and emphasised that individual sites retain their own ecological and land-use histories.

Changes in the revised article. The Methods now explain why proximity to a pollen site must be balanced against the number of archaeological dates available. Supplementary Figs. S35–S37 and Supplementary Table S9 present the 250 and 500 km comparison. The Discussion explains why it cannot substitute for site-specific pollen-source estimates.

Human events variable

Even by itself, the development of this variable represents a major undertaking and whilst I appreciate the substantial effort, I find the decomposition of continuous human change into categorical events represented by a binary variable overly simplistic and almost certain not to capture diverse changes across the PAPs considered here. I also find there to be a lack of description and justification of the method. Which experts were asked? Why were these experts

asked? Who do they represent? Why and how were these events chosen? By whom? What were the exact criteria used for each event * region considered? Indeed, for Central and South America, the authors use an established index from the literature, which underlines the extent of the work required for the development of indices such as these for individual continents, let alone for all continents as is attempted here.

The reductive nature of the index is exemplified best for North America, for which the authors provide three event categories: No Evidence of Impact, First Cultivation and European Settlement. For many Indigenous North American peoples, cultivating domesticated plant types remained supplementary to a much broader focus on foraging wild foods (both plant and animal) over the Holocene. Wild foods were often foraged with the substantial assistance of fire, which was used extensively across the continent before, during and after cultivation of domestic crops. According to this index, there was no human impact in North America before First cultivation, which is certainly untrue. Indigenous groups across North America have impacted vegetative communities over the entire Holocene and into the Pleistocene both directly, through burning, and indirectly, through the removal of the largest animals. Both of these would have strongly impacted vegetative communities, but their effects are entirely missed by the coarse categories here. It would be a vast amount of work to develop a pollen-based index of human impact for all regions globally that was transparent and not overly reductive. I do not believe that, in its present form, the inclusion of this binary variable adds to this work and I would simply remove it and curtail the analyses at the end of the period of SPD relevance, however defined.

Response. We agree that a binary event chronology cannot capture the continuity and diversity of past human land use. Clearance, grazing, cultivation, burning, and settlement varied in form, duration, intensity, and ecological consequence among regions. We have therefore removed “No Evidence of Impact” as an interpretation: absence of a recorded event means only that the specified pollen indicator was not identified. Nevertheless, we retain the chronology as secondary historical evidence because it brings locally interpreted indications of land use into a study otherwise dependent on broad archaeological SPDs, includes some younger observations, and allows us to ask whether the climate–human comparison changes when human activity is represented differently. Its partial dependence on the pollen record is stated explicitly, and it is not presented as independent confirmation.

The comparison is limited to places and periods for which an expert chronology was actually available: 151 pollen records in the spatial analysis and 61 continental-group–age combinations in the temporal analysis. Adding or substituting the event evidence changed estimates in both directions. This variability is historically informative because it shows that the inferred human contribution depends in part on the archive used to represent past activity. It does not extend the principal SPD conclusion beyond 2.0 ka BP. John Birks and Hilary Birks assessed Europe and North America; John Birks and Kuber Bhatta assessed Asia; and Simon Haberle and Annika Herbert assessed Oceania. They were chosen for regional palaeoecological and pollen-analytical expertise. The Central and South American

indicators follow the published regional synthesis. Supplementary Methods describes how the diagrams were assessed, and Supplementary Table S4 lists the regional event criteria and their limitations.

Changes in the revised article. The event chronology is now presented as a complementary historical perspective rather than evidence equal to the archaeological SPD. The Methods identify the experts and regional indicators, the Discussion addresses the incompleteness and partial circularity of the evidence, and the Supplementary Information compares results based on SPDs, events, and both sources together.

Statistical analyses

Firstly, I cannot see a summary of the total variance explained by the models here. This is of crucial importance because if the models explain a very small amount of the total variation in the data, then it would be difficult to say that humans/climate are most/not important drivers of the PAPs, in my view. These should be added as a Supplementary table and summarised in some way - perhaps across records, by region - as an Extended Data Figure.

Response. We agree that the strength of the ecological interpretation depends on how much variation the models explain. For the 1,098 pollen records in the principal spatial analysis, the mean adjusted R^2 is 0.440, the median is 0.450, and the middle half of the values lies between 0.326 and 0.567. Supplementary Fig. S23 shows how explanatory power varies among continental groups and regions, because a global average can conceal pollen records for which climate and human activity account for little of the observed history. Supplementary Table S5 separately reports the larger set of 1,156 records for which the original comparison of climatic and human contributions could be estimated.

During revision, we also corrected how the separate climatic and human contributions were extracted. The revised article retains the signed estimates, uses non-negative values only for the principal visual summary, and reports a further comparison that omits models with negative components. Climate had the larger estimated contribution under each treatment, although the numerical values differed. We stress that these contributions divide the variation explained by a model and must not be read as proportions of all change in the pollen assemblage.

Changes in the revised article. The Results now report overall model fit alongside the comparison of climate and human activity. Supplementary figures and tables show its geographical distribution, its relationship to the estimated contributions, and the effect of alternative treatments of shared or negative explanatory variation.

Secondly, temporal and spatial autocorrelation do not seem to be accounted for in these models, and their effects not checked for. These effects potentially confound model inferences and thus

either an explanation as to why they were not included needs to be added to the methods, or some additional variables need to be added to the models. This is likely of most importance to those analyses in Fig 2 where there will undoubtedly be a substantial trend component to the PAPs over the long period considered.

Response. We agree that ecological observations close together in time or space are not independent. Long-term succession and directional change can make neighbouring levels within a pollen sequence resemble one another, while shared geography can produce similar assemblages among nearby sites. The principal analysis of individual records now accounts for temporal structure, and the continental-group analysis through time accounts for spatial structure among sites. We also examine remaining spatial pattern in the residuals, repeat analyses using more widely spaced records, and test whether the broad result depends on any single continental group or region. These checks retain the general predominance of climate, while showing that some individual estimates remain uncertain or sensitive to analytical choices.

Changes in the revised article. Figs. 3 and 4 now present the analyses that account for temporal or spatial structure. The Methods describe these controls, and the Supplementary Information reports checks for remaining geographical pattern, the influence of closely spaced sites, and the effect of excluding individual continental groups or regions.

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Response. We thank the reviewer for these references. The revised article uses Giesecke et al. (2019), Gordon et al. (2024), and Mottl et al. (2021) to place the results within later-Holocene histories of pollen diversity, vegetation change, and land use. Chaput and Gajewski (2016), Crema (2022), and Crema and Bevan (2021) support the discussion of how radiocarbon dates can, and cannot, inform histories of human population.

Reviewer #3 (Remarks to the Author)

I work on Quaternary paleoclimate reconstructions and with Earth system scientists, and can comment on the relevance and impact of this study. It is indeed assumed by many that humans have played a significant and increasing role in shaping Holocene vegetation assemblage dynamics for the 8.5 to 2 ka BP study interval. This assumption is relevant, for instance, to the vigorous debate about whether the Anthropocene should be viewed as having gradually unfolded over tens of millennia (the “Anthropocene event” of Gibbard et al., 2022; see also the gradualist depiction in fig. 1 of Ellis, 2023), or whether significant impacts are restricted to more recent times. The conclusion that humans were not responsible for broad-scale ecological processes during this interval of study is therefore important to this debate, and it sets a baseline for human-driven impacts that were to later overwhelm Earth system functioning.

The study itself is based on the novel statistical analysis of two very large and independent data sets. The conclusions seem to be well supported by the data, and careful thought is given to explaining their limitations. There is one controversial aspect of Holocene climate that is not addressed but perhaps could be. It concerns claims by Ruddiman et al. (e.g. 2003, 2007, 2020) that the emergence of intensive rice cultivation in eastern Asia was responsible for the release of significant carbon dioxide and methane into the atmosphere, from 7 ka and 5 ka respectively. This would explain why global temperatures remain relatively flat even as summer insolation at 65°N declined from its peak in the Early Holocene. These claims have been questioned by Zalasiewicz et al. (2024) who noted that isotopic and modelling studies are consistent with oceanic outgassing rather than anthropogenic causes for these rises in greenhouse gases. Moreover, Summerhayes et al. (2024) noted that global average insolation may have in fact risen slightly over the past 7 kyr, potentially leading to the degassing of the global ocean which is by far the largest CO₂ reservoir on the planet. The question then, as to whether there is evidence within the pollen record for substantial cultivation of rice in eastern Asia, becomes relevant in assessing the extent of human impacts on the Earth system.

Response. We thank the reviewer for placing the study within the debate over early human influence on the Earth system. The history of rice cultivation in eastern Asia is clearly relevant to the early anthropogenic hypothesis. Our evidence, however, cannot resolve that question. The ten pollen assemblage properties describe broad changes in richness, evenness, and compositional reorganisation; they do not identify rice pollen or reconstruct the extent and intensity of cultivation. Nor do the present models examine greenhouse-gas sources. Consequently, the larger estimated climatic contribution in our analyses neither supports nor refutes the proposal that early rice agriculture affected atmospheric carbon dioxide or methane. A direct test would require harmonised taxon-level rice evidence, regional histories of cultivation and land use, and an analytical design linking those histories to greenhouse-gas change.

Changes in the revised article. The final Discussion paragraph now locates the result within the early anthropogenic debate while defining its evidential boundary. It explains why broad pollen assemblage properties cannot substitute for rice-pollen records, archaeological histories of cultivation, or evidence concerning the sources of Holocene greenhouse gases.

I consider the MS to be publishable after minor revision, and perhaps following consideration of the question of eastern Asia rice cultivation. The Holocene is crucial to our understanding of the rise of civilizations and the emergence of the modern world. This MS will have a profound impact on the way we approach these fundamental issues.

Martin J. Head

5 July 2025

Response. We thank the reviewer for this assessment. The revised article now discusses eastern-Asian rice cultivation as an important historical and Earth-system question while making clear that the present broad-scale pollen analysis cannot adjudicate between competing explanations of Holocene greenhouse-gas change.

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Response. We thank the reviewer for this literature. It is now used to frame the early anthropogenic debate and the alternative anthropogenic and oceanic explanations for Holocene greenhouse-gas change. We state explicitly that the evidence analysed here does not distinguish between those mechanisms.

Minor detailed comments on the MS

Line 69. Should there be consistency here in the precision, i.e. 8.5–2.0 ka BP? See Line 336 where this consistency is maintained.

Response. Agreed. We now use 8.5–2.0 ka BP consistently when specifying the study interval.

Line 82. For land use, reference to Our World in Data would be better: <https://ourworldindata.org/land-use>.

Response. Thank you for the suggestion. We added the Our World in Data synthesis to document the long-term expansion of cropland, grazing land, and settlements. Peer-reviewed studies remain the basis for causal interpretations and for the geological discussion of the Anthropocene.

Line 95. not “middle Holocene” but “Middle Holocene” if you are referring to the formal substage/subepoch (8.2–4.2 ka), or “mid-Holocene” if more loosely.

Response. Corrected. We use “Middle Holocene” for the formal subdivision and “mid-Holocene” for informal usage.

Line 113. Pleistocene and Holocene epochs (not periods).

Response. Corrected to “epochs.”

Line 116. “the term “period” not needed here because the Holocene can only refer to a unit of time.

Response. Corrected. We removed the unnecessary use of “period” with Holocene.

Line 124. reflects, not reflect (“each” refers to the singular).

Response. Corrected to “reflects.”

Line 134. “on the period between 8.5 and 2.0 ka BP” if the consistency in precision is justified.

Response. Corrected, with the study interval consistently expressed as 8.5–2.0 ka BP or “between 8.5 and 2.0 ka BP,” according to sentence structure.

Line 147. we would expect an altered covariance between the PAPs within different climate zones on continental regions.

Response. Corrected and clarified. H2 is now expressed as a prediction about historical changes in the relationships among pollen assemblage properties within continental groups and their regions, rather than as a claim about the response of particular plant taxa.

Line 150. This subheading is a little confusing because while it references H1, it describes the result rather than the stated hypothesis. I realize that restating the hypothesis could be misleading here, so perhaps frame the hypothesis as “H1: Holocene palaeoclimate versus human impact”.

Response. Agreed. We revised the subheading to “H1: Holocene human impact secondary to palaeoclimate,” thereby linking the hypothesis to its ecological and temporal setting while indicating the observed direction of the result.

Line 157. Middle to Late Holocene – you are referring to official subseries of the Holocene, so the first initial of the modifiers “middle” and “late” must be capitalized. If this is not done, you will need to define what will then be informal terms.

Response. Corrected to “Middle to Late Holocene” where the formal subdivisions are intended.

Line 175. Again an issue of consistency in precision. In Asia 3.5 ka BP so for Europe should that be 3.0 ka BP?

Response. Corrected. Ages are now reported with consistent precision within each comparison.

Line 218. palaeoclimate data between 8.5 and 2.0 ka BP to investigate ... (then delete “8.5 and 2.0 ka BP” at the end of the sentence).

Response. Corrected and the duplicated interval was removed.

Line 220. Our first hypothesis posited that ...

Response. Corrected. The sentence now begins with an explicit formulation of the first hypothesis.

Line 241. PAPs not PAPS.

Response. Corrected throughout to “PAPs.”

Line 275. “mid” is not a stand-alone word in most dictionaries, so it would be “mid-Holocene” as an informal interval of time.

Response. Corrected to “mid-Holocene.”

Line 326. “factors like” better as “such factors as”.

Response. Corrected to “such factors as.”

Line 336. “from 8.5 to 2.0 ka BP”. Use an en-dash when referring to, but not specifying, a range, such as “over the range 8.5–2.0 ka BP”. I note here that “2.0 ka unlike elsewhere is states to two significant figures, which seems appropriate precision if supported by the dates themselves.

Response. Corrected. We use “from 8.5 to 2.0 ka BP” in prose and an en dash in the adjectival or compact range “8.5–2.0 ka BP.”

Line 467. You do make it clear that Eric C Grimm is no longer with us, but are you perhaps dedicating this MS to the memory of the person rather than the person themselves? Remember to change “manuscript” to “article” when revising the MS.

Response. Corrected. The article is now dedicated to the memory of Eric C. Grimm, and submission-facing prose refers to the revised work as an article.